

The OTA Requestor MAY choose to attempt a [QueryImage](#) command on a different OTA Provider in its OTA Provider List to determine if an update is available from that other OTA Provider.

Otherwise, if there are no other OTA Providers available to query, the OTA Requestor SHALL NOT re-query the OTA Provider which was the subject of the command sooner than the longest of either:

- 2 minutes (120 seconds) from the last [QueryImage](#) command;
- the delay stated by the [DelayedActionTime](#) field, if present.

Note that if a Node loses its timekeeping state due to events such as power loss or restart, the above timing constraint MAY be ignored, however, the previously stated overriding constraint of a minimum delay of 120 seconds between queries to any single OTA Provider by a given OTA Requestor have to be respected.

Handling errors from QueryImage Command

If an OTA Requestor hits error conditions of any kind in invoking the QueryImage command, including receiving DownloadProtocolNotSupported in Status, there are two possible outcomes:

1. If the OTA Requestor has a Software Image it had previously successfully downloaded and verified, the OTA Requestor SHOULD skip the Query step, and move directly to the Apply step (see [Section 11.20.3.6, “Applying a software update”](#)). This increases the likelihood that the OTA Requestor will eventually succeed to apply a previously transferred Software Image.
2. If the OTA Requestor is still attempting to discover an OTA Update, it MAY choose to attempt a [QueryImage](#) command on a different OTA Provider in its OTA Provider List, in which case the timing for the query SHALL match the query timing constraints expressed in the previous paragraphs of this section. Otherwise, it SHALL continue to query the same OTA Provider, again following the query timing constraints previously expressed.

11.20.3.3. Availability of Software Images

The algorithm used by the Image Selection Logic to determine availability of a new Software Image SHALL consider all fields provided by the OTA Requestor and attempt to provide the newest matching Software Image. The OTA Image Selection Logic SHALL only provide newer (numerically higher) SoftwareVersion than the SoftwareVersion sent in the query. See [Section 11.20.3.3.2, “Conceptual algorithm for matching OTA Software Images applicable to a query”](#) for more details.

The OTA Provider MAY provide a Software Image that only conveys data for a subset of updateable components within the OTA Requestor’s Node. These cases of partial or componentized software updates are determined purely by the entity generating the OTA Software Image, and the OTA Provider SHALL never mutate the contents of an OTA Software Image.

The original provider of a Software Image SHOULD be able to assume the contents of the Software Image will remain unchanged and signatures would remain valid. Therefore, an OTA Provider SHALL NOT modify the contents of any Software Images other than allowing that OTA Requestor to index into the Software Image using the [BDX Protocol](#) or other supported download protocol, such that the OTA Requestor may obtain only the desired parts of the Software Image.

The OTA Provider SHALL NOT hide or otherwise mask the contents of a Software Image available